NEURAL SYSTEMS EXECUTIVE SUMMARY

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The research uses two deep-learning architectures, Artificial Neural Network (ANN/MLP) and Convolutional Neural Network (CNN), to identify Fashion-MNIST photos (28x28 grayscale, 10 apparel categories). Comparing their accuracy, generalization, and performance gains via regularization and data augmentation is the aim.

Dataset Overview:

Fashion-MNIST dataset (10,000 test and 60,000 training pictures)
T-shirt/top, trousers, pullover, dress, coat, sandal, shirt, sneaker, bag, ankle Boot
Each grayscale image has 28 by 28 pixels, making it an easy benchmark for visual classification.

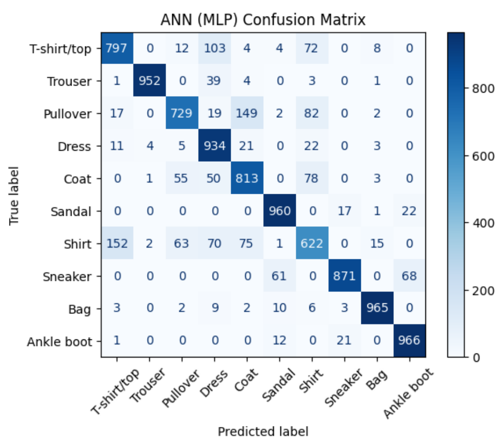
Model Architectures:

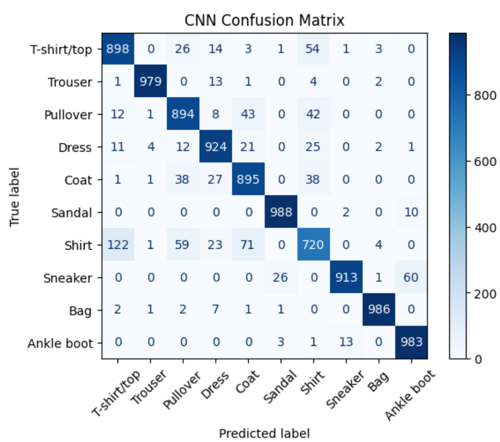
First. ANN (MLP): Dropout → Dense(64) → Flatten → Dense(128) → Dropout → Dense(10)
Reduced dropout rate and optional batch normalization are improvements.
Lacking spatial awareness, it captures global patterns.
Two. CNN: Conv(32) → MaxPool → Conv(64) → MaxPool → Flatten → Dense(128) → Dropout → Dense(10)
Convolution layers were added, data augmentation was moderate, and dropout was 0.3.
Higher accuracy is achieved by learning local spatial hierarchies.

Training Setup
Optimizer: Adam
Loss Function: Categorical Cross-Entropy
Batch Size: 128
Epochs: 20–30
Callbacks: Early stopping (patience=10), learning rate reduction (optional)
Data Augmentation: Applied only to CNN (rotations ±10°, small translations, no flips).

Performance Summary:

Model Performance: • ANN (MLP): 86.1% (upgraded), 86.09% (basic)
• CNN: 92.0% (upgraded), 91.80% (basic)
Because CNN can capture spatial relationships, it consistently outperformed ANN.
Improved models had less overfitting, smoother validation curves, and improved generalization.





Error & Visualization Analysis:

Frequent misclassifications between visually similar items were emphasized via confusion matrices:
T-shirt/top; shirt; pullover; coat
• Ankle boot and sneaker
Training curves showed CNN converged faster and achieved higher validation stability than MLP.

Conclusions:

CNN-based models performed better when it came to classifying images. Both accuracy and generalization were enhanced by improvements like dropout, batch normalization, and augmentation. The majority of the remaining misunderstanding was between classes that were visually similar.

Recommendations:

- To improve feature extraction, increase CNN's depth and filter count.
- To increase prediction robustness, investigate ensemble approaches.
- Make use of learning-rate schedulers and powerful optimizers (AdamW, RAdam).
- For improved explainability, incorporate model interpretability techniques (such as Grad-CAM).